

PROJECT REPORT

AI Smart Waste Segregation Dustbin

AI-Based Smart Waste Segregation Dustbin Using ESP32-CAM, Edge Impulse, Arduino Uno, Ultrasonic Sensor, and Servo Motor

Domain: Artificial Intelligence · Embedded Systems · IoT

Technology Stack: ESP32-CAM · Edge Impulse (Tiny ML) · Arduino Uno

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1. Project Title

AI-Based Smart Waste Segregation Dustbin Using ESP32-CAM, Edge Impulse, Arduino Uno, Ultrasonic Sensor, and Servo Motor

2. Project Objective

The objective of this project is to automatically identify whether waste is Reusable or Non-Reusable using Artificial Intelligence (AI). After classification, the dustbin automatically rotates its lid or sorting mechanism using a servo motor so that the waste falls into the correct compartment.

This project reduces manual waste segregation, improves recycling efficiency, and demonstrates the integration of AI, Embedded Systems, and IoT.

3. Problem Statement

Waste segregation is often done manually, which is:

- Time-consuming
- Error-prone
- Unhygienic
- Inefficient for recycling

To solve this problem, an AI-powered smart dustbin automatically identifies the type of waste and separates it into the correct bin.

4. Components Used

Hardware

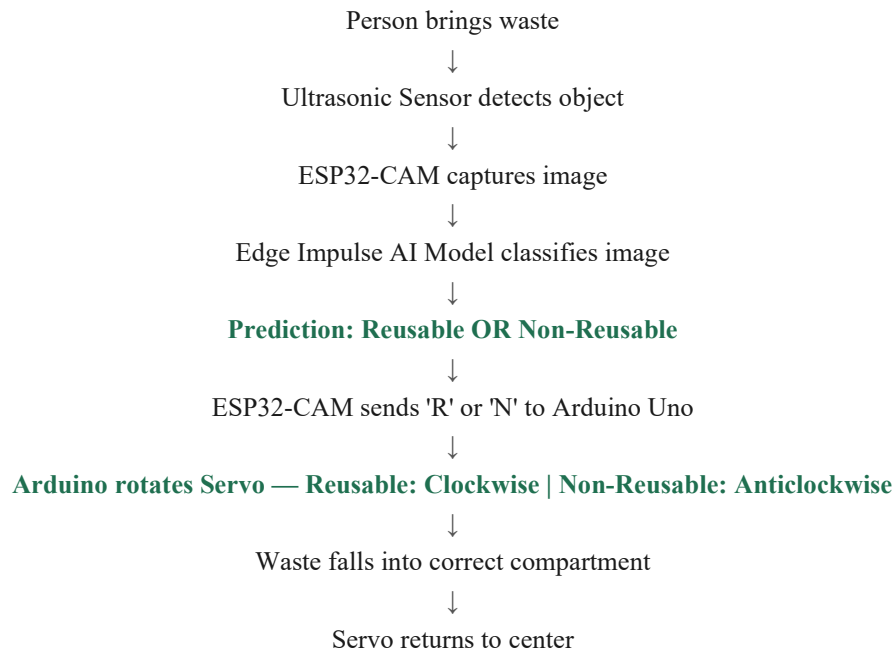
- ESP32-CAM
- Arduino Uno
- HC-SR04 Ultrasonic Sensor
- SG90 Servo Motor
- Breadboard
- Jumper Wires
- USB Cable
- 5V Power Supply

Software

- Arduino IDE
- Edge Impulse
- ESP32 Board Package
- C++
- GitHub

5. Working Principle

The complete working process is as follows:



6. AI Model

The AI model was developed using Edge Impulse.

Dataset

Two classes were created:

Reusable — Examples:

- Plastic Bottle
- Paper
- Glass
- Wood
- Rubber

Non-Reusable — Examples:

- Banana Peel
- Food Waste
- Vegetable Waste
- Organic Waste

Approximately 50 images per class were uploaded and used for training.

7. Image Processing

- The ESP32-CAM captures a real-time image.
- The image is sent to the Edge Impulse model running locally on the ESP32-CAM.
- The model outputs class probabilities, for example: Reusable: 0.98, Non-Reusable: 0.02
- The class with the higher probability is selected.

8. Communication

After prediction:

- If Reusable probability > Non-Reusable probability → ESP32 sends 'R'
- Otherwise → ESP32 sends 'N'

This data is transmitted to the Arduino Uno using UART serial communication.

9. Arduino Function

The Arduino continuously performs the following:

- Reads ultrasonic distance.
- Waits until an object is detected.
- Receives 'R' or 'N' from the ESP32.
- Rotates the servo accordingly.

Servo Movement

Classification	Servo Action
Reusable	Servo moves to 180°, then returns to 90°
Non-Reusable	Servo moves to 0°, then returns to 90°

10. Ultrasonic Sensor

Purpose: Avoid unnecessary image processing. The camera captures an image only when an object is placed near the dustbin.

Distance Threshold: 15 cm

11. Servo Motor

The servo controls the sorting flap.

- Clockwise Rotation → Reusable Waste
- Anticlockwise Rotation → Non-Reusable Waste

After 2 seconds, the servo returns to 90°.

12. Technologies Used

- Artificial Intelligence
- Tiny Machine Learning (TinyML)
- Embedded Systems
- Image Classification
- UART Communication
- Arduino Programming
- ESP32 Development
- Edge Impulse

13. Advantages

- Automatic waste segregation
- Reduces human effort
- Improves recycling
- Low-cost implementation
- Compact design
- Fast classification
- Runs offline (no cloud required)
- Real-time operation

14. Limitations

- Works only for trained waste categories.
- Classification accuracy depends on lighting and camera angle.
- Cannot identify unknown objects reliably.
- Small dataset can reduce accuracy.

15. Future Improvements

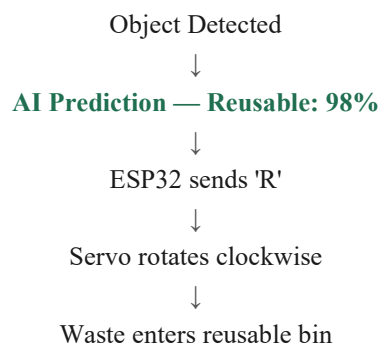
- Add more waste categories (metal, e-waste, hazardous waste).
- Cloud monitoring using IoT.
- Mobile application for monitoring.
- OLED/LCD display to show classification results.
- Automatic bin level monitoring.
- Voice feedback.
- Battery-powered portable version.
- Higher accuracy with a larger dataset.

16. Skills Demonstrated

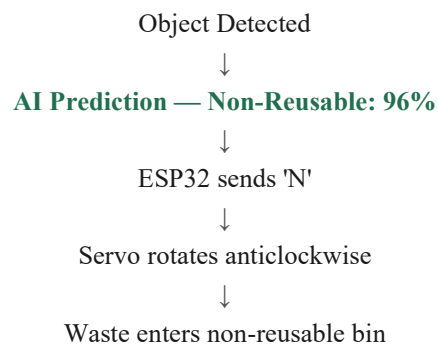
- Embedded Systems
- Arduino Programming
- ESP32-CAM Development
- TinyML
- Edge AI
- Image Classification
- Sensor Interfacing
- Servo Motor Control
- UART Communication
- Hardware Integration
- IoT Fundamentals
- GitHub Project Management

17. Expected Output

Reusable Item (Plastic Bottle)



Non-Reusable Item (Banana Peel)



18. Conclusion

The AI Smart Waste Segregation Dustbin combines Artificial Intelligence, Computer Vision, and Embedded Systems to automate waste classification and sorting. Using an ESP32-CAM running an Edge Impulse image classification model, the system identifies whether an item is Reusable or Non-Reusable. The result is sent to an Arduino Uno, which uses an ultrasonic sensor to detect when an object is present and controls a servo motor to direct the waste into the appropriate compartment.

The project demonstrates practical skills in TinyML, AI deployment on edge devices, sensor integration, microcontroller programming, and real-time embedded system design. It is an excellent portfolio project for electronics, embedded systems, IoT, and AI-focused internships and job applications.